Topic: Fuzzy testing
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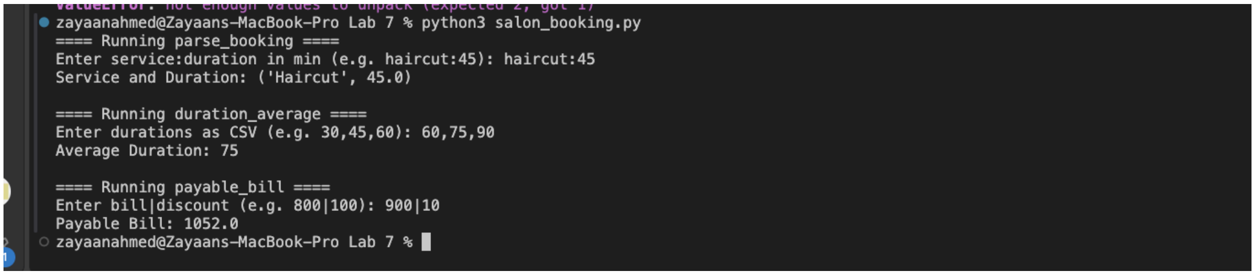
Terminal Screenshot after Running Faulty salon_booking.py:

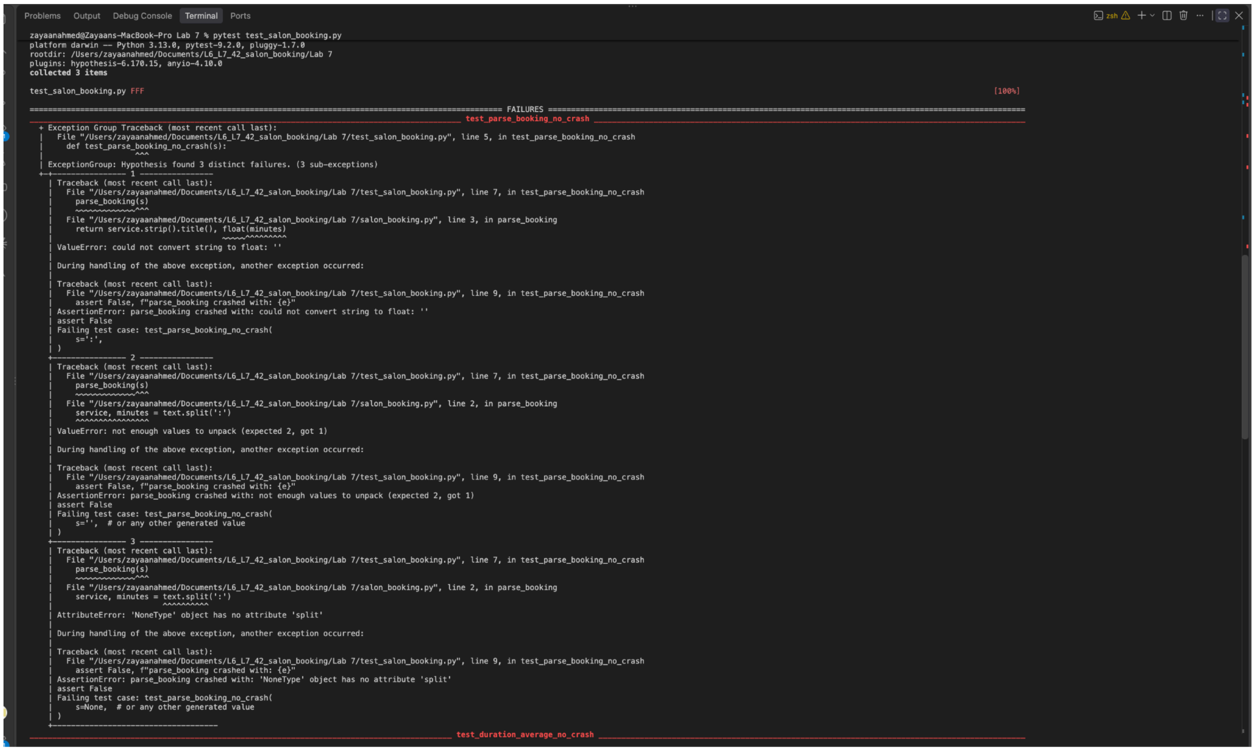


```
ValueError: not enough values to unpack (expected 2, got 1)
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % python3 salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut:45
Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 60,75,90
Average Duration: 75

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 900|10
Payable Bill: 1052.0
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % 
```

Terminal Screenshot after updating test_salon_booking.py:



```
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % pytest test_salon_booking.py
platform darwin -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7
plugins: hypothesis-6.170.15, anyio-4.10.0
collected 3 items

test_salon_booking.py FFF                                                                                    [100%]

=============================================== FAILURES ===============================================
_______________________________________ test_parse_booking_no_crash ____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 5, in test_parse_booking_no_crash
  |     def test_parse_booking_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 7, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 3, in parse_booking
    |     return service.strip().title(), float(minutes)
    |                                     ~~~~~^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 9, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed with: {e}"
    | AssertionError: parse_booking crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 7, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 2, in parse_booking
    |     service, minutes = text.split(':')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 9, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed with: {e}"
    | AssertionError: parse_booking crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 7, in test_parse_booking_no_crash
    |     parse_booking(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 2, in parse_booking
    |     service, minutes = text.split(':')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 9, in test_parse_booking_no_crash
    |     assert False, f"parse_booking crashed with: {e}"
    | AssertionError: parse_booking crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_booking_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_______________________________________ test_duration_average_no_crash _________________________________
```

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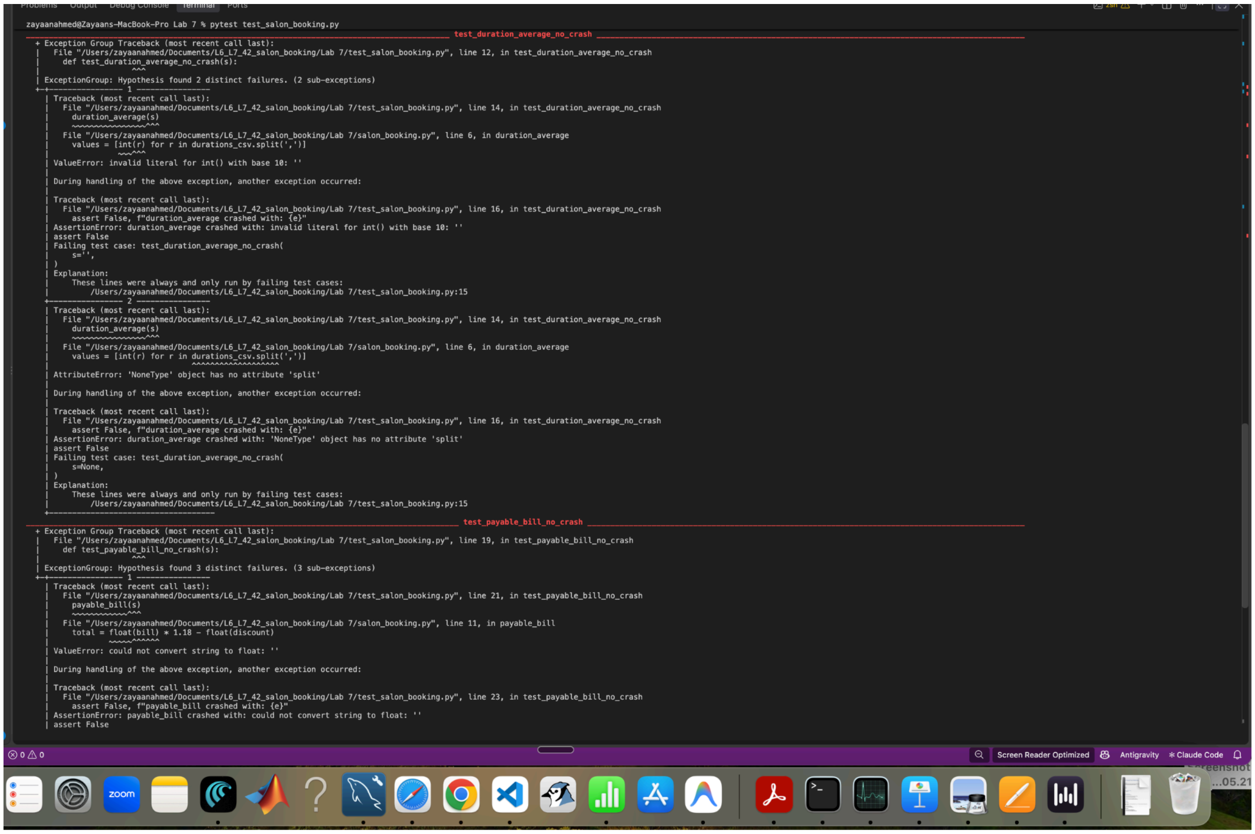
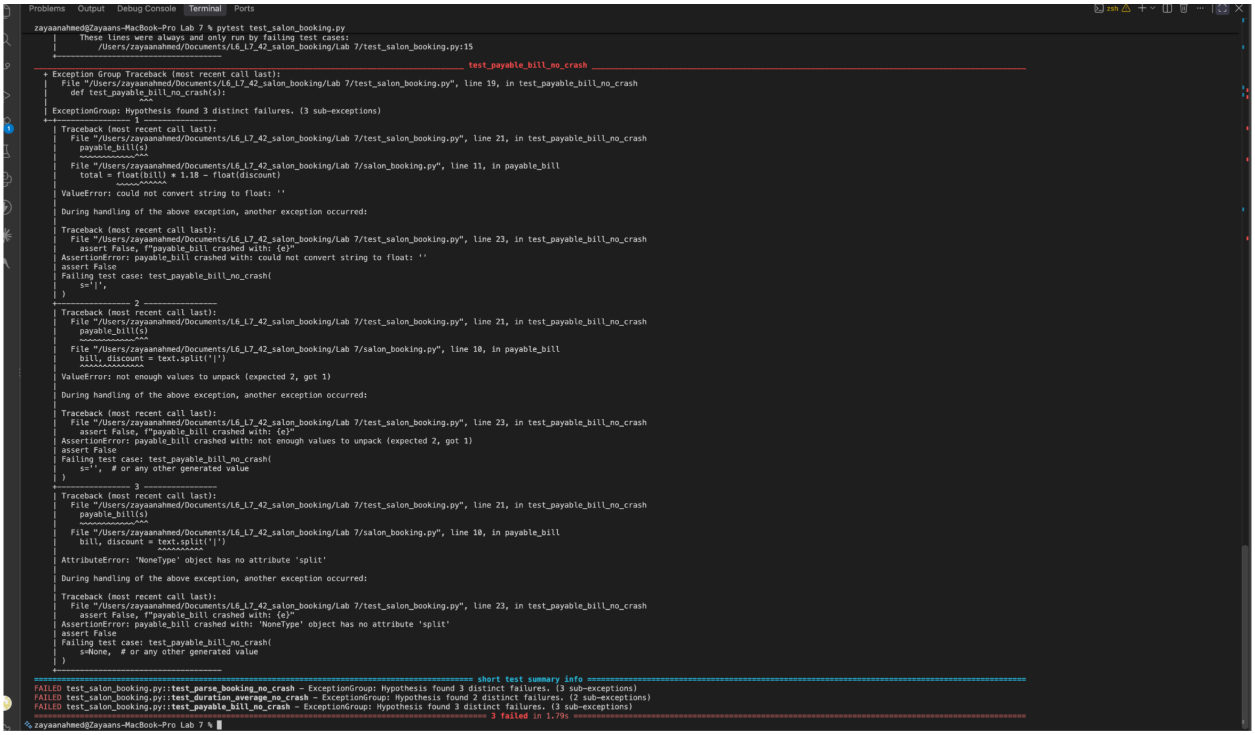
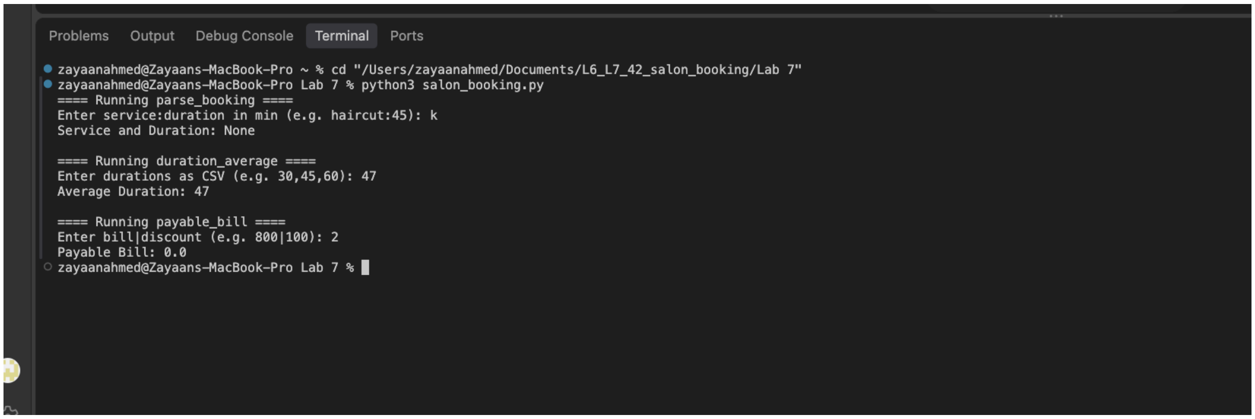


```
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % pytest test_salon_booking.py
______________________________________________________ test_duration_average_no_crash ______________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 12, in test_duration_average_no_crash
  |     def test_duration_average_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 14, in test_duration_average_no_crash
    |     duration_average(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 6, in duration_average
    |     values = [int(r) for r in durations_csv.split(',')]
    |               ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 16, in test_duration_average_no_crash
    |     assert False, f"duration_average crashed with: {e}"
    | AssertionError: duration_average crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_duration_average_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py:15
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 14, in test_duration_average_no_crash
    |     duration_average(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 6, in duration_average
    |     values = [int(r) for r in durations_csv.split(',')]
    |                               ^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 16, in test_duration_average_no_crash
    |     assert False, f"duration_average crashed with: {e}"
    | AssertionError: duration_average crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_duration_average_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py:15
    +------------------------------------
_______________________________________________________ test_payable_bill_no_crash _______________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 19, in test_payable_bill_no_crash
  |     def test_payable_bill_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 21, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 11, in payable_bill
    |     total = float(bill) * 1.18 - float(discount)
    |             ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 23, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e}"
    | AssertionError: payable_bill crashed with: could not convert string to float: ''
    | assert False
```



```
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % pytest test_salon_booking.py
    |     These lines were always and only run by failing test cases:
    |         /Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py:15
    +------------------------------------
_______________________________________________________ test_payable_bill_no_crash _______________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 19, in test_payable_bill_no_crash
  |     def test_payable_bill_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 21, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 11, in payable_bill
    |     total = float(bill) * 1.18 - float(discount)
    |             ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 23, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e}"
    | AssertionError: payable_bill crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 21, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 23, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e}"
    | AssertionError: payable_bill crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 21, in test_payable_bill_no_crash
    |     payable_bill(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/salon_booking.py", line 10, in payable_bill
    |     bill, discount = text.split('|')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7/test_salon_booking.py", line 23, in test_payable_bill_no_crash
    |     assert False, f"payable_bill crashed with: {e}"
    | AssertionError: payable_bill crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_payable_bill_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
================================================ short test summary info ================================================
FAILED test_salon_booking.py::test_parse_booking_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_salon_booking.py::test_duration_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_salon_booking.py::test_payable_bill_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
==================================================== 3 failed in 1.79s ====================================================
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % 
```

Terminal Screenshot After Handling Edge Cases in salon_booking.py and running it:

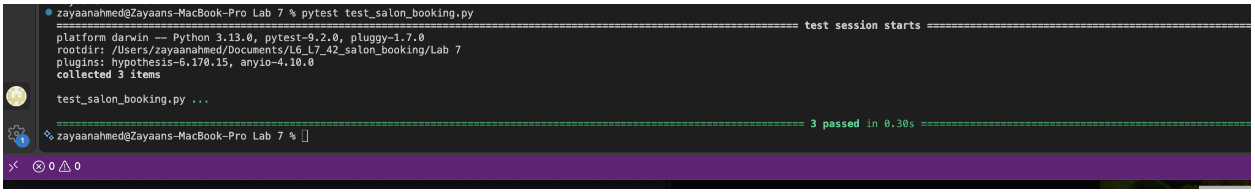


```
zayaanahmed@Zayaans-MacBook-Pro ~ % cd "/Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7"
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % python3 salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): k
Service and Duration: None

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 47
Average Duration: 47

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 2
Payable Bill: 0.0
zayaanahmed@Zayaans-MacBook-Pro Lab 7 %
```

Terminal Screenshot After Handling Edge Case and running test_salon_booking.py:



```
zayaanahmed@Zayaans-MacBook-Pro Lab 7 % pytest test_salon_booking.py
========================================================================================================================= test session starts ==============================================================
platform darwin -- Python 3.13.0, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/zayaanahmed/Documents/L6_L7_42_salon_booking/Lab 7
plugins: hypothesis-6.170.15, anyio-4.10.0
collected 3 items

test_salon_booking.py ...

========================================================================================================================== 3 passed in 0.30s ==============================================================
zayaanahmed@Zayaans-MacBook-Pro Lab 7 %
```